Portalwiz® Technologies | BUSINESS SOLUTIONS

Computer Vision & Microservice Backend (Document AI)

Position	Intern
Submission	GitHub repository

Instructions

Build a RESTful backend API using Python (FastAPI or Flask) that accepts an image of an ID card (e.g., Driver's License or Employee Badge), detects the document boundary, applies basic image preprocessing, uses an OCR engine to extract key information, and returns structured JSON output.

Key Requirements & Scope

1. Document Detection & Preprocessing (OpenCV)

- **Accept image uploads via a POST endpoint.**
- **Crop or align the ROI (Region of Interest) containing the document card.**
- **Apply basic image processing to optimize OCR readability (e.g., grayscale, adaptive thresholding, or noise reduction).**

2. OCR & Information Extraction (Python AI/ML Stack)

- **Use an open-source OCR library (such as `pytesseract`, `easyocr`, or `PaddleOCR`).**
- **Extract specific fields using regular expressions or heuristic parsing:**
  - **Full Name**
  - **Document / ID Number**
  - **Date of Birth / Expiry Date**
- **Extract the face thumbnail from the ID photo region and save it to disk.**

3. Backend API & Storage

- **Endpoint 1 (`POST /api/v1/process-id`): Accepts an image file, processes it, and returns extracted fields along with image metadata.**
- **Endpoint 2 (`GET /api/v1/records/{id}`): Retrieves previously processed results from a local database (SQLite/PostgreSQL) or JSON store.**
- **Basic input validation and HTTP error handling (e.g., handling invalid file formats or unreadable images).**

4. Technical Stack Requirements

- **Language: Python 3.9+**
- **Framework: FastAPI or Flask**
- **CV/ML Tools: OpenCV, EasyOCR / Pytesseract**
- **Storage: SQLite or File-based JSON**